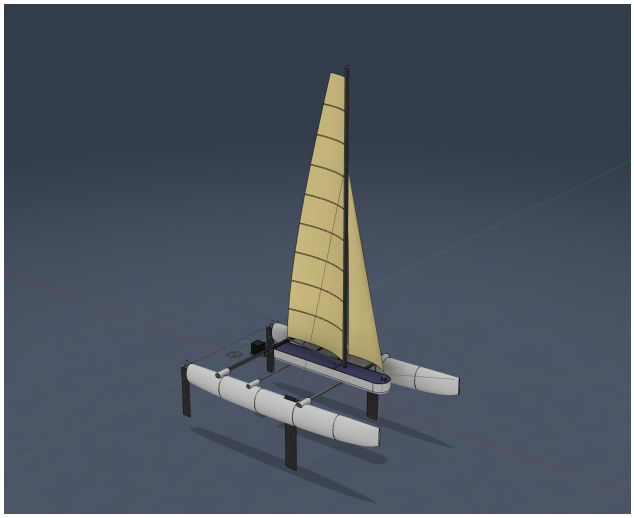
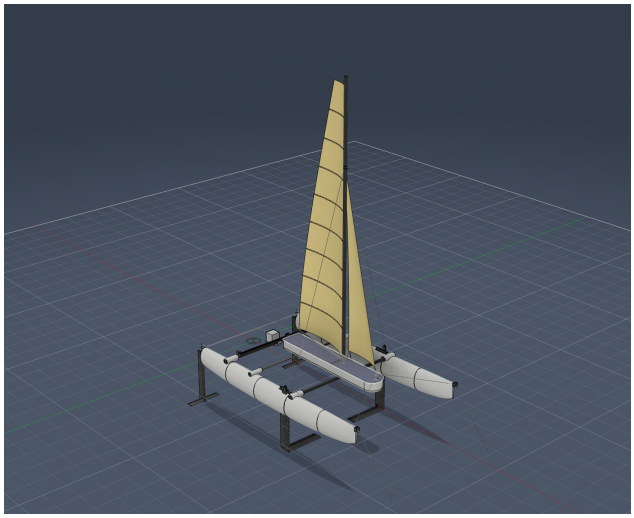
# RC Catamaran 1 m: Build Manual

Build-first boat and the v2.3 foiling kit

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September 2026





## 1 What you are building

A 1 m catamaran with a beach-cat rig: a square-top, fully battened main on a rotating wing mast, a jib on a forestay bridle, and three sail controls (jib sheet, traveler, mainsheet tension) on three servos. Build the boat with daggerboards first (the *build-first* version, 2114 g ready to sail). Once it sails well, the foiling kit replaces the daggerboards with L-foils, adds wand-driven flaps for ride-height control, and puts T-foil elevators on the rudders (2373 g).

Predicted, not yet measured: the build-first boat does 2.65 m/s on a beam reach in 8 kn of wind. The foiling version flies from about 7.1 kn on a beam reach and 8.9 kn at 60° to the wind, and does 4.53 m/s on a beam reach in 12 kn.

**Build the 0.6 m Mini Cat first** (its own manual). Many steps here assume you have done them once at that size.

## 2 What it costs

Build-first boat **$381.02**; foiling kit **$101.17** more. Prices were read on retailer pages in September 2026 (sources at the end). Shipping, tax, a NiMH charger, the printer and hand tools (including a 0.9 mm hex key for the elevator trim screws) are not included. The FS-i6X radio and 4xAA pack from the Mini Cat carry over.

| Group | Item | Use | Qty | Total |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| **Build-first boat** | | | | |
| Radio | FlySky FS-i6X + FS-iA6B receiver | 4 channels used: rudder, traveler, jib, mainsheet | 1 | $69.99 |
| Radio | AA alkaline x4 | transmitter power | 1 | $2.99 |
| Servos | JX W4505-4T sail winch servo | traveler (about 231 mm of car travel, rig model) | 1 | $11.13 |
| Servos | JX W4505-2T sail winch servo | jib sheet (about 93 mm of sheet, rig model) | 1 | $11.13 |
| Servos | Feetech FS5115M standard servo | mainsheet tension | 1 | $32.06 |
| Servos | MG90S metal-gear micro servo | rudders via pushrod + tie bar | 1 | $2.59 |
| Power | Panasonic eneloop AA 2000 mAh, 4-pack | receiver pack (4.8 V) | 1 | $12.75 |
| Power | Traxxas 4-cell AA holder with switch (TRA1523) | receiver battery box + on/off | 1 | $4.00 |
| Printed | PETG filament 1 kg | hulls, pod, lid, rudder-servo bracket and cover, rudders, boards, gudgeons, tracks, cars, mast fairing and fittings ( 1.5 kg used, 1 m manual print list) | 2 | $27.98 |
| Carbon | Carbon tube 16 x 14 mm x 1 m | two 600 mm crossbeams | 2 | $32.56 |
| Carbon | Carbon tube 12 x 10 mm x 1 m | two-piece mast (1.3 m) | 2 | $24.52 |
| Carbon | Carbon tube 10 x 8 mm x 1 m | mast joiner spigot (150 mm) and pod beam (600 mm) | 1 | $16.15 |
| Carbon | Carbon tube 8 x 7 mm x 1 m | boom (345 mm) | 1 | $17.20 |
| Carbon | Carbon rod 1.2 mm x 2 m | 8 battens | 1 | $5.25 |
| Hardware | Music wire 0.156 in | rudder stocks, tie bar | 2 | $9.98 |
| Hardware | Music wire 0.093 in | rudder pushrod, track end-stop turning pins, gooseneck pin, boom sheet eye | 1 | $3.99 |
| Hardware | Du-Bro micro pushrod system | clevises | 1 | $8.55 |
| Sails | Icarex PC31 ripstop polyester, 1 m | main + jib (0.40 m2) | 1 | $21.99 |
| Sails | Ripstop sail tape 2 in, half roll | luff, batten pockets, corners | 1 | $24.00 |
| Rigging | Braided PE line 30 lb | stays, bridle, sheets | 1 | $14.43 |
| Adhesives | BSI 5-minute epoxy 9 oz | hull joints, beams, fittings | 1 | $22.99 |
| Adhesives | BSI gap-filling CA 1/2 oz | small parts | 1 | $4.79 |
| **Foiling kit** | | | | |
| Printed | PETG filament 1 kg | L-foil, elevator and flap cores, trim pods, ride-height linkage ( 290 g, 1 m manual print list) | 1 | $13.99 |
| Carbon | Unidirectional carbon tape 2 in x 12 ft | spar caps: 4 layers per face on struts, wings, elevators | 2 | $39.90 |
| Adhesives | EL2 laminating epoxy 500 g | wet-out of the caps | 1 | $20.70 |
| Carbon | Carbon rod 1.2 mm x 1 m | flap hinge pins | 1 | $3.47 |
| Carbon | Carbon rod 2 mm x 1 m | two 300 mm ride-height wands, elevator pivot pins, wand and bellcrank axles | 1 | $8.85 |
| Hardware | Music wire 0.047 in | wand-to-flap pushrods | 1 | $3.99 |
| Hardware | Du-Bro micro pushrod system | clevises for the flap linkage | 1 | $8.55 |

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|---|---|---|---|---|
| Hardware | M2 x 6 mm stainless set screw, cup point (Bolt Depot 8697) | elevator trim screws, two per trim pod | 4 | $1.72 |

# 3 Printing

PETG throughout. The hull and pod segments are real 0.8 mm skins with the bulkheads built in: slice them with 2 perimeters and 0 % infill, so each segment prints as a sealed compartment. Print the rudder blades, daggerboards, foil cores, flaps and trim pods solid (100 % infill is our recommendation for stiffness, not a tested setting). Filament below is for solid parts at 1.27 g/cm$^3$ and for the skins as modelled.

| STL file (folder `stl/`) | Qty | Size (mm) | PETG (g) |
|---|---|---|---|
| **Boat** | | | |
| `Hull_Port_Seg_1_x0_200.stl` | 1 | 200 x 117 x 82 | 69 |
| `Hull_Port_Seg_2_x200_400.stl` | 1 | 200 x 120 x 84 | 80 |
| `Hull_Port_Seg_3_x400_600.stl` | 1 | 200 x 117 x 81 | 91 |
| `Hull_Port_Seg_4_x600_800.stl` | 1 | 200 x 118 x 83 | 78 |
| `Hull_Port_Seg_5_x800_1000.stl` | 1 | 200 x 115 x 65 | 54 |
| `Hull_Stbd_Seg_1_x0_200.stl` | 1 | 200 x 117 x 82 | 69 |
| `Hull_Stbd_Seg_2_x200_400.stl` | 1 | 200 x 120 x 84 | 80 |
| `Hull_Stbd_Seg_3_x400_600.stl` | 1 | 200 x 117 x 81 | 91 |
| `Hull_Stbd_Seg_4_x600_800.stl` | 1 | 200 x 118 x 83 | 78 |
| `Hull_Stbd_Seg_5_x800_1000.stl` | 1 | 200 x 115 x 65 | 54 |
| `Pod_Lid_Seg_1_x270_380.stl` | 1 | 110 x 90 x 1 | 15 |
| `Pod_Lid_Seg_2_x380_600.stl` | 1 | 220 x 90 x 1 | 30 |
| `Pod_Lid_Seg_3_x600_840.stl` | 1 | 240 x 90 x 11 | 34 |
| `Pod_Shell_Seg_1_x270_380.stl` | 1 | 110 x 90 x 57 | 47 |
| `Pod_Shell_Seg_2_x380_600.stl` | 1 | 220 x 90 x 54 | 92 |
| `Pod_Shell_Seg_3_x600_840.stl` | 1 | 240 x 90 x 60 | 79 |
| `Rudder_Servo_Bracket.stl` | 1 | 73 x 46 x 26 | 23 |
| `Rudder_Servo_Cover.stl` | 1 | 45 x 44 x 43 | 14 |
| `Rudder_Blade_x2.stl` | 2 | 226 x 45 x 5 | 96 |
| `Daggerboard_x2.stl` | 2 | 325 x 60 x 7 | 244 |
| `Gudgeon_x2.stl` | 2 | 28 x 24 x 16 | 10 |
| `Traveler_Track_Centre.stl` | 1 | 152 x 7 x 6 | 6 |
| `Traveler_Track_Port.stl` | 1 | 158 x 25 x 11 | 8 |
| `Traveler_Track_Stbd.stl` | 1 | 158 x 25 x 11 | 8 |
| `Traveler_Car.stl` | 1 | 18 x 16 x 11 | 2 |
| `Jib_Track.stl` | 1 | 96 x 10 x 5 | 3 |
| `Jib_Car.stl` | 1 | 14 x 14 x 5 | 1 |
| `Bridle_Post.stl` | 1 | 23 x 7 x 7 | 1 |
| `Mast_Cap.stl` | 1 | 14 x 14 x 14 | 2 |
| `Hounds_Collar.stl` | 1 | 16 x 16 x 12 | 1 |
| `Mast_Fairing_Lower_x5.stl` | 5 | 171 x 24 x 14 | 29 |
| `Mast_Fairing_Upper_x2.stl` | 2 | 182 x 24 x 14 | 12 |
| `Gooseneck.stl` | 1 | 32 x 16 x 14 | 3 |
| `Boom_Jaw.stl` | 1 | 28 x 8 x 8 | 2 |
| **Foiling kit** | | | |
| `Foil_Core_L_Port.stl` | 1 | 296 x 184 x 60 | 136 |
| `Foil_Core_L_Stbd.stl` | 1 | 296 x 184 x 60 | 136 |
| `Foil_Core_Elevator_x2.stl` | 2 | 160 x 28 x 3 | 18 |
| `Foil_Flap_x2.stl` | 2 | 165 x 12 x 3 | 9 |

| | | | |
|---|---|---|---|
| Elevator_Trim_Pod_x2.stl | 2 | 45 x 14 x 9 | 7 |
| Foiling_Rudder_Extension_x2.stl | 2 | 45 x 8 x 5 | 3 |
| Wand_Hinge_Bracket_x2.stl | 2 | 17 x 17 x 13 | 4 |
| Wand_Hub_x2.stl | 2 | 23 x 16 x 12 | 3 |
| Bellcrank_Mount_x2.stl | 2 | 37 x 24 x 10 | 6 |
| Strut_Bellcrank_x2.stl | 2 | 12 x 12 x 2 | 0 |
| Flap_Horn_x2.stl | 2 | 7 x 3 x 3 | 0 |

Total: about 1508 g for the boat and 323 g for the foiling kit, before supports and failed prints.
Print the seven mast fairing pieces in spiral-vase mode: one 0.45 mm perimeter, no bottom or top layers. They are thin shells that slide over the mast tube. Print the tracks and cars standing on end, so the T-profile lies flat on the bed. The linkage parts marked x2 are handed: print the second one mirrored, for the starboard hull.

# 4 Building the boat

## 1. Hulls

1. Print the ten hull segments. Each ends in half of a bulkhead; the faces meet flat.
2. Epoxy the segments of each hull face to face in order (x = 0 is the transom), keeping the keel line straight on a flat surface.
3. Seal test: submerge each hull in a tub for 10 minutes and look for bubbles or water. Each segment is a separate compartment, so one leak floods only one.

## 2. Crossbeams and pod beam

1. Cut two 600 mm crossbeams from the 16 x 14 mm tube, and a 600 mm pod beam plus the 150 mm mast joiner from the 10 x 8 mm tube (wear a mask when cutting or sanding carbon).
2. Epoxy the crossbeams into the front and rear sockets of both hulls and the pod beam into the third socket, with the hulls parallel and the beams square.

## 3. Pod and electronics

1. Bond the three pod shell segments and the three lid segments. The pod snaps onto the front beam and the pod beam.
2. Fit the two sail winches (traveler: 4-turn; jib sheet: 2-turn) and the mainsheet servo on their rails. The winches hang by their ears in 4–6 mm wells. JX does not publish the winch's ear height; the rails use the FS5115M value (27.3 mm above the case bottom). If a winch case touches the bottom of its well first, shim under the ears.
3. Hook the mainsheet tail to the servo horn 25 mm from the shaft, not closer. The sheet runs twice between the traveler car and the boom, so a shorter arm lets the servo pull harder on the beams and mast than the structure check allows. Fit a longer horn if the supplied ones are shorter.
4. Receiver and the 4xAA pack with its switch go in the pod. The two ends of the traveler line and the mainsheet tail leave the pod through the three holes in its transom and run aft to the turning pins in the traveler track's end stops (Step 7).

## 4. Rudders

1. Rudder stocks from the 0.156 in music wire, through the printed blades (pivot at 25 % chord).
2. Epoxy a gudgeon to each transom, centred on the hull, with its plate flat on the transom face; its two lugs hold the stock 19 mm behind the transom. Slide the stock down through both lugs into the blade; the tiller arm above the upper lug keeps it in.
3. Snap the rudder-servo bracket onto the rear beam from below, centred between the hulls, and epoxy it so it cannot rotate. The MG90S hangs by its ears on the bracket's rails, aft of the beam, under the printed cover. The pushrod (0.093 in wire, Du-Bro clevises) runs from the slot in the cover to the tie bar, which joins the two tiller arms.

## 5. Rig

1. Mast: the two 12 x 10 mm tubes joined by the 150 mm spigot, 1298 mm overall. Plug the top with the mast cap (epoxy), so the tube stays sealed if the boat capsizes.
2. Wing: from the foot up, the gooseneck (its fork and pin hole aft), the five lower fairing pieces, the hounds collar with its centre 928 mm above the mast foot, and the two upper pieces. Line up the trailing edges and fix each piece to the tube with a drop of CA at both ends. The whole mast turns in the pod lid and the step, so the wing follows the sail.
3. Melt 1.2 mm lacing holes through the fairing 4 mm ahead of its trailing edge, every 50 mm, starting 50 mm above the sail's foot; the marks on the sail pattern give the heights. Skip the hounds collar.
4. Standing rigging from the 30 lb braided line (lengths between fittings): forestay 873 mm, each bridle leg 336 mm, each shroud 974 mm. Add your own allowance for knots. Tie the three stays to a loop of line in the hounds collar's groove, so the mast turns inside the loop. At the bridle post, the bridle line runs through the lower hole (fix it there with CA); the forestay and the jib tack tie through the upper hole.
5. Boom: 328 mm of the 8 x 7 mm tube, with the boom jaw epoxied into its front end. Put the jaw's tongue between the gooseneck's cheeks and pin it with a piece of the 0.093 in wire. Bend a small eye from the same wire for the mainsheet and epoxy it under the boom's aft end.
6. Traveler track: epoxy the centre piece on top of the rear beam, centred. Slide the car on from one end, then epoxy the two end pieces on so the rail runs continuously, and sand the joints smooth. Press a 19 mm piece of the 0.093 in wire into each end stop's pin hole. The car stops 213 mm either side of the centre line (boom 35° out).
7. Jib track: slide the jib car on, then epoxy the track on the pod lid on the centre line, 590 mm forward of the transoms. Drill a 2 mm hole through the lid under the hole at the track's centre, for the jib sheet.

## 6. Sails

1. Print the 1:1 patterns (`manual_1m/sail_patterns_1m_1to1.pdf`, a single tall sheet: a copy shop's plotter, or tile it) and check the 100 mm scale bar.
2. Cut the main (foot 345, luff 1190, head 105 mm) and the jib (foot 221, luff 849 mm) from the Icarex. Tape the edges, corners and eight batten pockets with the sail tape.
3. Battens from the 1.2 mm carbon rod, bottom to top (mm): 331, 318, 300, 276, 249, 216, 180, 140.

4. Main luff: tape it, then melt the lacing holes 5 mm in from the edge at the marks on the pattern. Lace it to the wing's trailing edge with the braided line, one turn round the edge per hole. Lash the tack to the hole in the boom jaw and the head to the top lacing hole.
5. Jib: fold a 10 mm sleeve along the luff around the forestay, as on the Mini Cat.

### 7. Mainsheet routing

The traveler car moves up to 213 mm either side of the centre line. That is far more line than the mainsheet servo can take up, so the car must not pull on the mainsheet. Tie the mainsheet to the port end-stop pin in the upper slot. Run it through the car's port post, up through the eye under the boom and back down, through the starboard post, round the starboard pin in the upper slot and forward into the pod. Moving the car then shortens one side of the sheet exactly as much as it lengthens the other.

The traveler line is one loop. From the winch drum it runs out of the pod, round the port pin in the lower slot, through the car's lower hole, round the starboard pin and back into the pod to the drum. Fix the car on the line with a drop of CA through the hole in the top of the car.

### 8. Radio setup

1. Bind the receiver and assign four channels: rudder, traveler, jib sheet, mainsheet tension.
2. Traveler winch: set its end points so the car runs the full track (about 209 mm of car travel in the rig model) without hitting the ends. Jib winch: about 82 mm of sheet from sheeted in to eased. Mainsheet servo: set its end points by eye, flattest leech without stalling the servo.
3. Do not sail with the jib sheeted fully in while the main is eased past about 30°: the jib sheet then crosses the boom (CAD clearance sweep). Ease the jib a little first.
4. Rudder: stick right, bow right. The linkage gives ±18° of rudder.
5. Failsafe: rudders centred and sheets eased on signal loss.

### 9. Balance and float test

Rigged, the build-first boat should balance 450 mm forward of the transoms (move the battery to correct it). Float it and check both hulls sit level; leave it 10 minutes and check the pod and hulls for water.

## 5 The foiling kit

Fit the kit only after the build-first boat sails well.

### 10. Foil cores and carbon caps

1. Print the L-foil cores (port and starboard are mirror images), the elevator cores and the flaps. The cores have 0.5 mm deep, 25 mm wide recesses on both faces of the strut, the wing and the elevator for the carbon caps.
2. Split the 2 in unidirectional tape lengthwise into 25 mm strips and lay 4 layers into each recess with the laminating epoxy, fibres along the span; the caps finish flush with the section. The caps use about 5.0 m of the 7.3 m of tape.
3. The strut has a 2.4 mm channel for the flap pushrod (0.047 in wire); keep it clear of epoxy.

## 11. Flaps

The flap is the aft 25 % of each wing (12 mm), from 10 mm inside the tip to the strut. It hinges on 1.2 mm carbon pins on its leading edge, 38 mm behind the wing's leading edge. The CAD does not model the hinge holes: drill them on that line at each end of the flap. The flap must swing ±15° freely.

## 12. Ride-height linkage

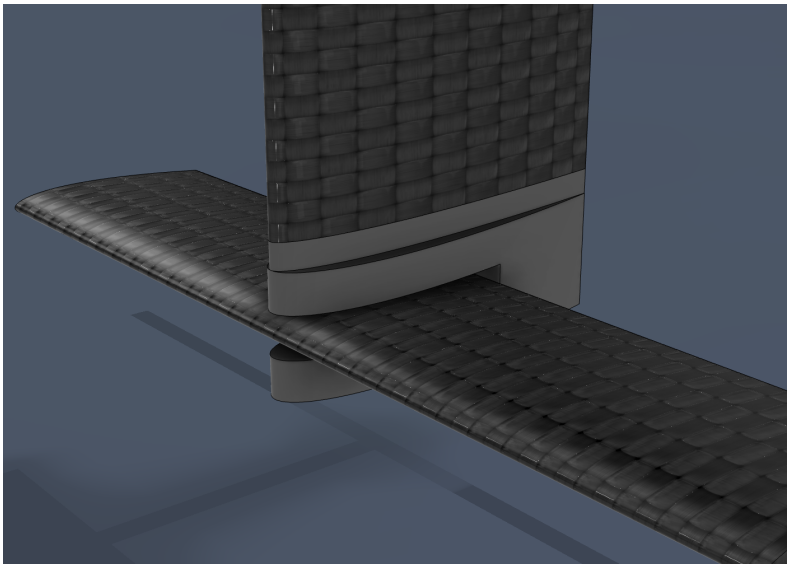
One wand per hull, 300 mm of 2 mm carbon rod, hinged at the bow. It drives its flap through a deck pushrod, a bellcrank and the pushrod inside the strut. The linkage must give:

- **zero flap** with the wand 49° below the deck line (the design ride height, hinge about 226 mm above the water);
- **1.5° of flap per degree of wand**, wand down giving flap down;
- the full ±15° of flap without binding.

The printed linkage gives them. The wand hub's horn is 8.25 mm long, the bellcrank's arms are 8 mm, and the flap horn's hole is 5.5 mm behind the flap hinge: $(8.25/5.5) \times (8/8) = 1.5°$ of flap per degree of wand.

1. Epoxy a wand hinge bracket to each stem, cheeks forward. Put the wand hub between the cheeks on a 2 mm carbon axle and push the wand into the hub's socket.
2. Epoxy the bellcrank mount on the deck beside the strut's top. Pin the bellcrank between its cheeks on a 2 mm carbon axle, one arm up and one arm aft.
3. Bond the flap horn to the upper face of the flap's root, in line with the slot in the strut's inboard face.
4. Bend Z-ends on the 0.047 in wire. The deck pushrod runs from the hub's horn to the bellcrank's upper arm; the strut pushrod runs from the bellcrank's aft arm down the strut's channel and out of the slot to the flap horn. Bend the last Z with the wand at 49° and the flap at zero.

## 13. Elevators and trim pods



1. Bond a 8 mm extension and then a trim pod under each rudder blade; the pod's slot is at the elevator's height.
2. Slide the elevator through the slot and pin it at its quarter chord with a 10 mm piece of the 2 mm carbon rod.
3. Two M2 set screws, one from above and one from below, clamp it 12 mm behind the pin. One turn moves the elevator about 1.9°. Set −0.5° (nose down) to start.

### 14. Fit and set up

1. Replace the daggerboards with the L-foils in the same trunks. Main wing incidence +1.5°.
2. Rebalance: the foiling boat should balance 454 mm forward of the transoms, 2373 g in total.
3. Run the bench checks and the tow test in `testing/RC_Foiling_Cat_Tow_Test_Plan.pdf` before sailing on the foils.
4. First flights: take off on a beam reach, where the foils have enough roll authority, then head up. Near take-off the leeward hull may keep touching; that is expected.

## Price sources (read September 2026)

- HobbyTown: `https://www.hobbytown.com/flysky-i6x-6channel-afhds-2a-air-radio-system-w-ia6b-receiver-fsy-i6xia6b/p1674181`
- HobbyTown (EcoPower AA x4 add-on): `https://www.hobbytown.com/flysky-fsgt2e-2channel-transmitter-w-fsa3-receiver-fsy-fs-gt2ea3/p1592822`
- AliExpress: listed $5.79 at 48% off; full price used: `https://www.aliexpress.com/item/1005003265131874.html`
- Pololu: `https://www.pololu.com/product/3426`
- Waveshare: `https://www.waveshare.com/mg90s-servo.htm`
- MicroBattery: `https://www.microbattery.com/panasonic-bk-3mcca4ba-eneloop-aa-2100-cycle-ni-mh-pre-charged-rechargeable-batteries-4-pack80.html`
- HobbyTown: `https://www.hobbytown.com/traxxas-4cell-aa-battery-holder-w-switch-cover-tra1523/p49945`
- Elegoo US: `https://us.elegoo.com/products/rapid-petg-filament-1-75mm-colored-1kg`
- Wind Catcher RC (price from search listing, page not reachable from here): `https://windcatcherrc.com/product/carbon-fiber-tube-16mm-x-14mm-x-1000mm/`
- RMRC: `https://www.readymaderc.com/products/details/carbon-fiber-tube-12mm-x-10mm-1m-long`
- CST CS-525 1M: `https://www.cstsales.com/Carbon_Fiber_Tubes.html`
- CST CS-316 2M: `https://www.cstsales.com/carbon_rods.html`
- K&S 0.156 in x 12 in: `https://ksmetals.com/collections/12-long-music-wire`
- HobbyTown: `https://www.hobbytown.com/dubro-pushrod-system-micro-dub847/p29360`
- Jones Airfoils, per metre (138 cm wide): `https://jonesairfoils.com/products/icarex`
- kiteboarding.com: $15.00 base + $9.00 half roll: `https://kiteboarding.com/proddetail.asp?prod=ripstop_2inch`
- Walmart, PowerPro 30 lb x 150 yd (read 2026-09-24; the original eBay listing had ended): `https://www.walmart.com/ip/PowerPro-Braided-Fishing-Line-30lb-x-150yd-ylw-21100300150Y/17117880`

- HobbyTown: https://www.hobbytown.com/bob-smith-industries-quickcure-5-minute-epoxy-9o-bsi202/p37082
- HobbyTown: https://www.hobbytown.com/bob-smith-industries-instacure-gap-filling-ca-1-2oz-bsi106/p37055
- CST CU-379: 2 in unidirectional carbon, 0.005 in, 12 ft: https://www.cstsales.com/heat_tack.html
- Easy Composites EL2, 500 g kit: https://www.easycomposites.us/el2-epoxy-laminating-resin
- Bolt Depot, per piece (read 2026-09-25): https://boltdepot.com/Product-Details?product=8697